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**MODULE: INTRODUCTION TO NETWORKING**

**ASSIGNMENT TITLE: NETWORK DESIGN USING CISCO-PACKETTRACER**

**DATE SUBMITTED: 6/2/2026**

**CASE STUDY:**

You have been hired as a junior network technician to design a small office network for a company called Databridge innovations.

The office has three departments:
1. Admin Department - handles internal operations
2 .Sales Department - handles customer communication.
3. HR Department - handles recruitment tasks.

The company wants the network designed and tested using Cisco Packet Tracer, ensuringthat: All devices are correctly configured and connected , and the departments must beableto communicate with each other using simple network test to confirmthe basicconnectivity.

**Project Expectation:**

Using the same network built:

1. Apply one simple ACL rule on the router.

2. Configure the ACL so that:

   a .The HR department (192.168.30.0/24) cannot ping the Admin department(192/168.10.0/24)

   b . Failed ping from HR to the Admin department.

Capture relevant screenshots placed at the appropriate points in the report. Capture the screenshot of the ACL command used.



**Objective/Executive Summary**

The aim of this project is to be able to design a small multi-department network,configureand connect networks using a Cisco Packet Tracer. It also involves applyingAccessControlLists ( ACLs ) on the router to manage and control network traffic between departments.This configuration is to preserve network security and integrity by ensuring a properaccessfor authorized departments by allowing Sales department to access the Admindepartmentwhile the HR department is denied access to the Admin department.

## Network Design

This network diagram shows a small organization network with three departments: HR,Salesand Admin Department.

The three switches(departments) were interconnected through a central router calledtheData-bridge and 2pcs were assigned to each department.The network setup involves;

1. Adding 1 Router (2911), 3 Switches (2960) and 6Pcs(2 in each department) inCisco-packet Tracer.

2. The router is renamed as the name of the company- Data-bridge . 3. The switches are renamed as the name of the department- Admin ,Sales andHR. 4. IP addresses were assigned to each switches which also serves as the gateway. 5. The Server and PCs were connected to the Switch (Admin_Dept) usingthestraight through Ethernet cable.

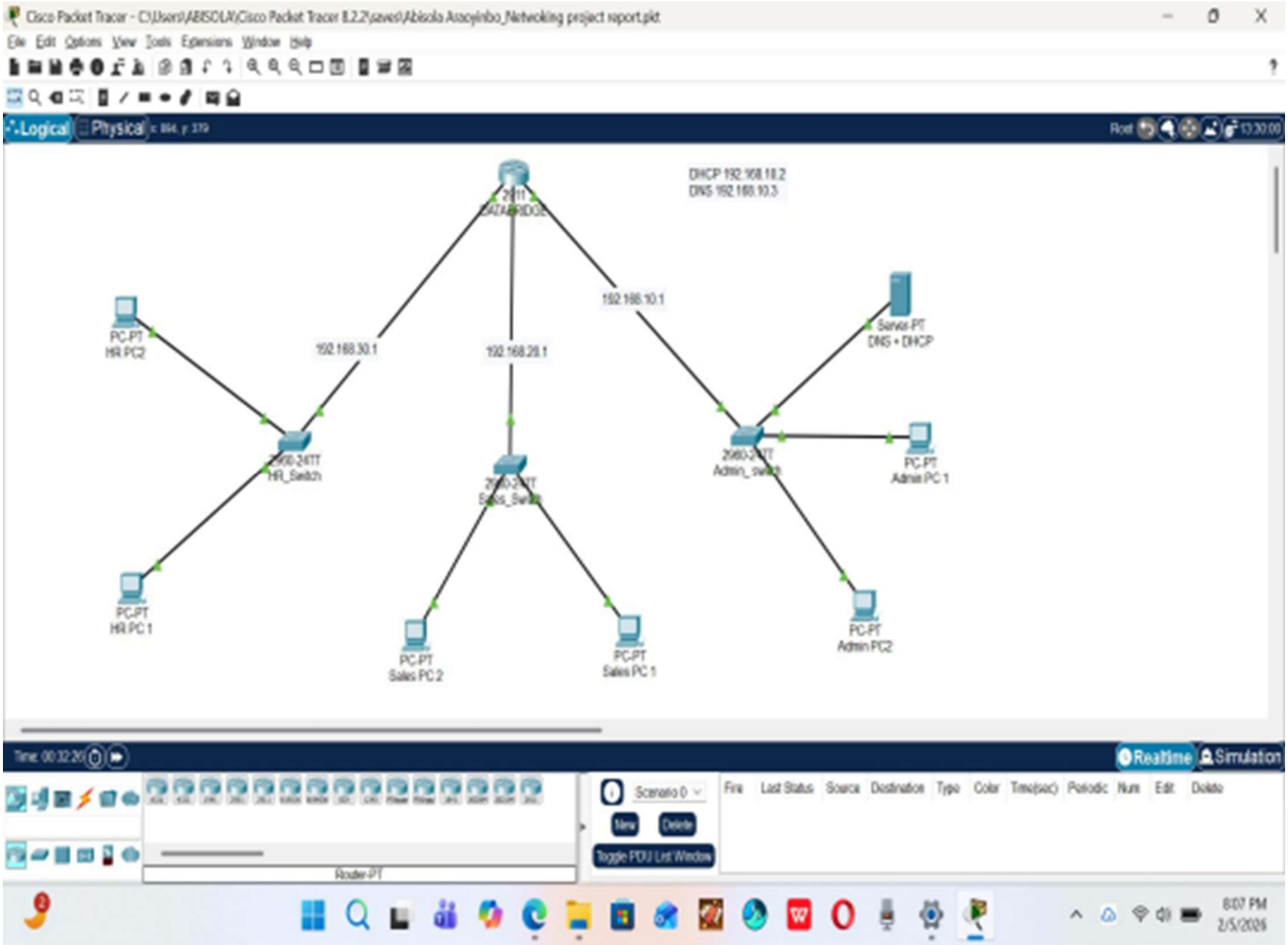
6. 1 server for DHCP and DNS .

7. The configuration for each switches were configured statically. 8. The 2 PCs in sales department are connected to the switch(Admin_switch) 9. The 2 PCs in the HR_dept are connected to the switch (HR_Dept) usingtheautomaticcable.

10.The switches Admin_Dept, Sales_dept, HR_dept are connected to the Router (Data-bridge using the automatic Ethernet cable.



Figure 1: A NETWORK DESIGN FOR DATABRIDGE




**Network Setup**:

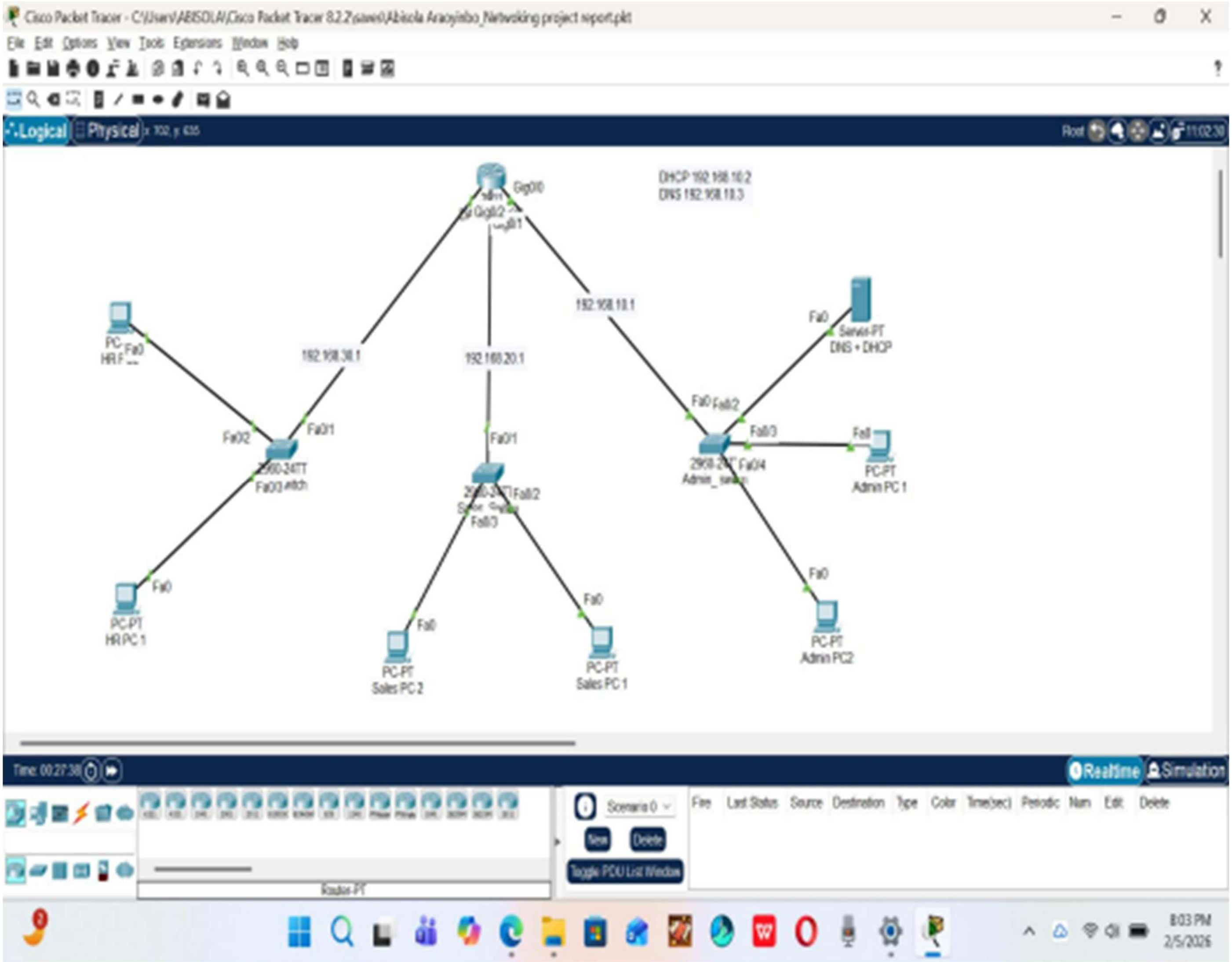
Router: The databrige router acts as the main network link connecting the threedepartmental subnets.

Admin Dept: Connected through the Admin-switch by using the 192.168.10.0/24subnet.HR Dept: Connected through the HR_switch by using the 192.168.30.0/24subnet. Sales Dept: Connected through the Sales Switch by using 192.168.20.0/24subnet.



The DHCP server ( 192.168.10.2) assigns IP addresses automatically to all PCsinalldepartment while the DNS server (192.168.10.3) ensures domain name translationintoIPaddresses.

Figure 2: The network diagram setup with port labels shown for Databridge.




**Configuration steps** : The router and server were configured using the CommandLineInterface(CLI) mode.



The CLI was enabled and set to global configuration mode by using the command“configureterminal”

The router interface was renamed to Databridge using the command “hostnameDatabridge”.

Figure 3: This diagram shows the configuration steps for the Router(data-bridge) andserver.

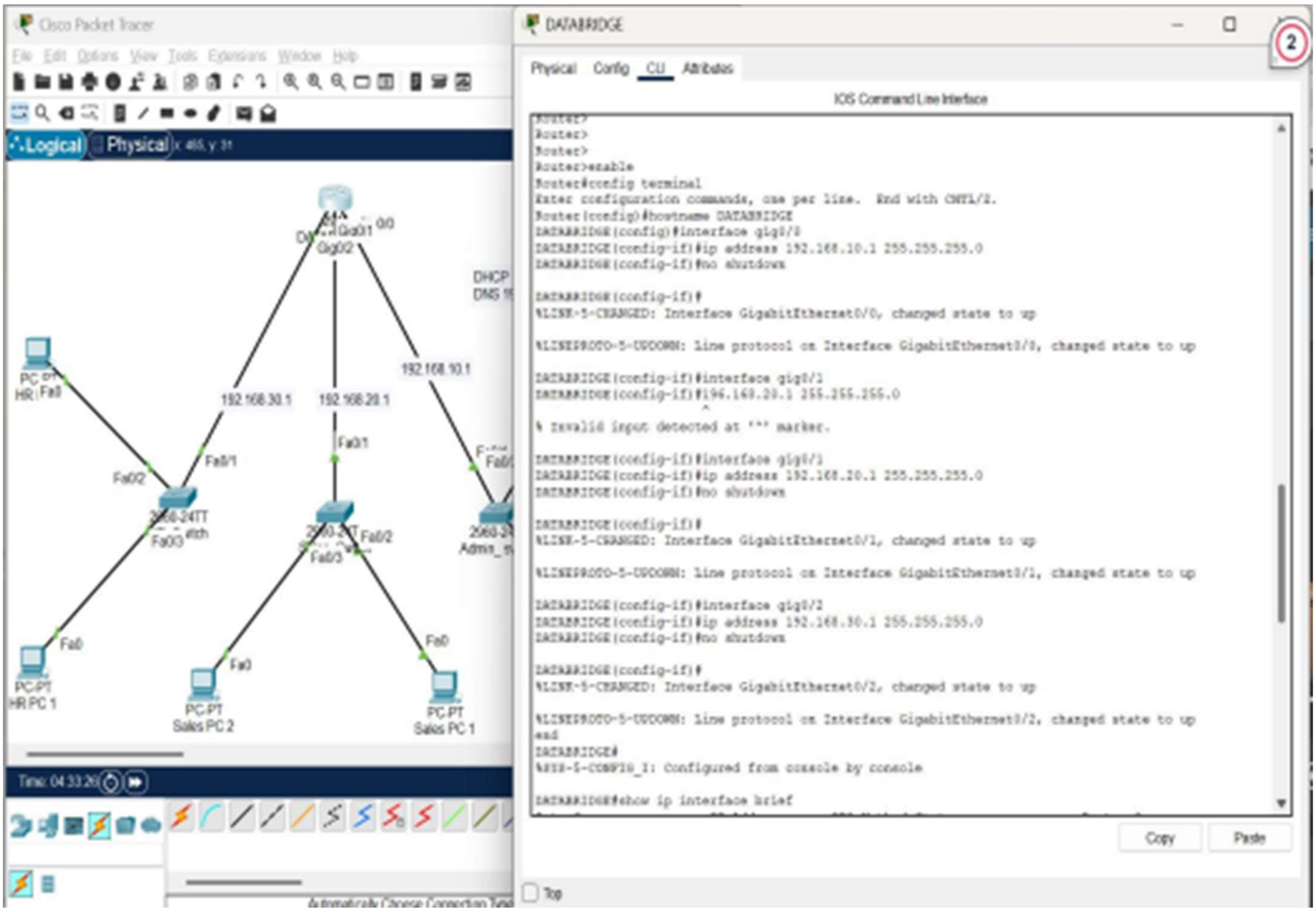
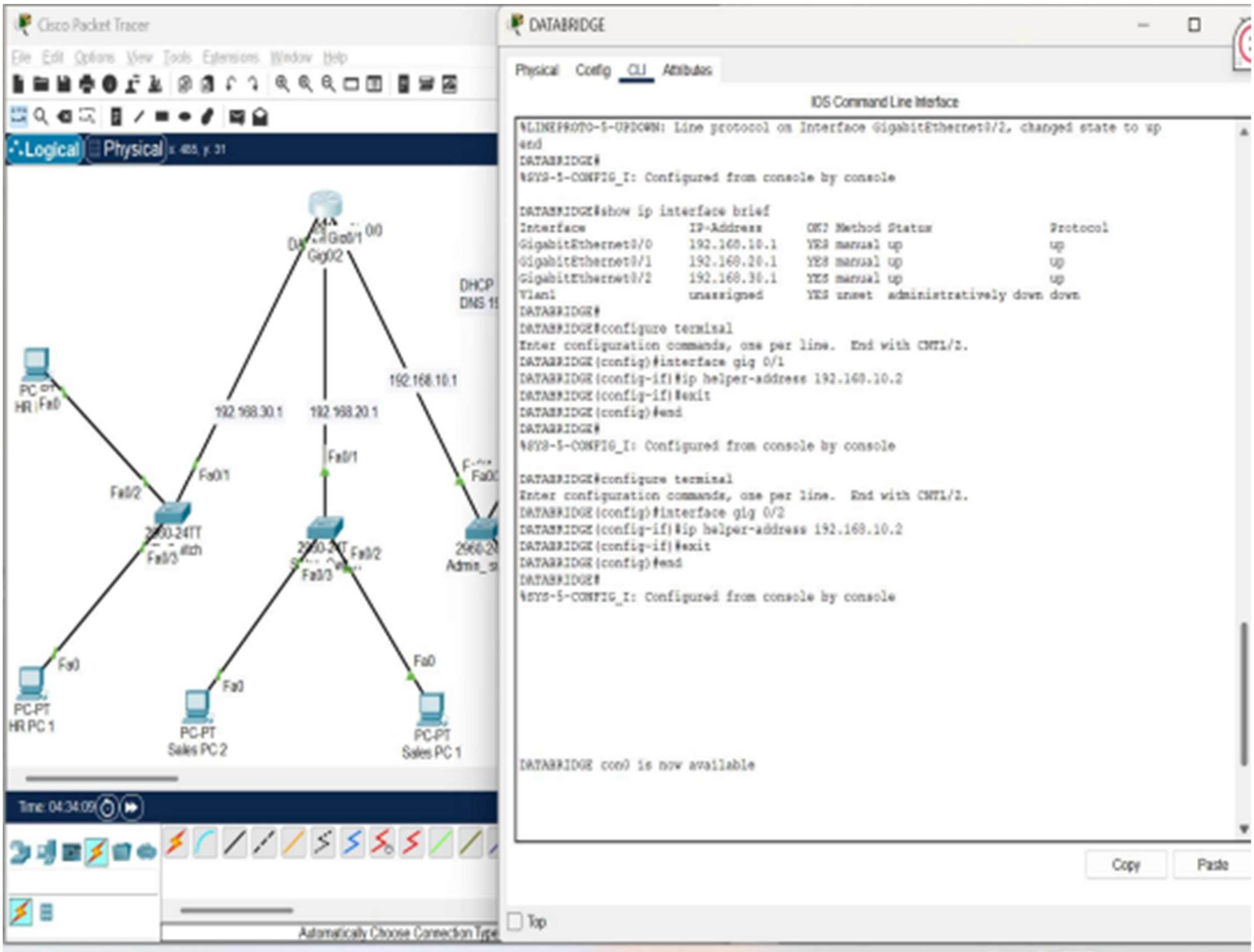





Figure 4: This shows the verification of the configured interfaces and the configurationoftheDHCP interface.

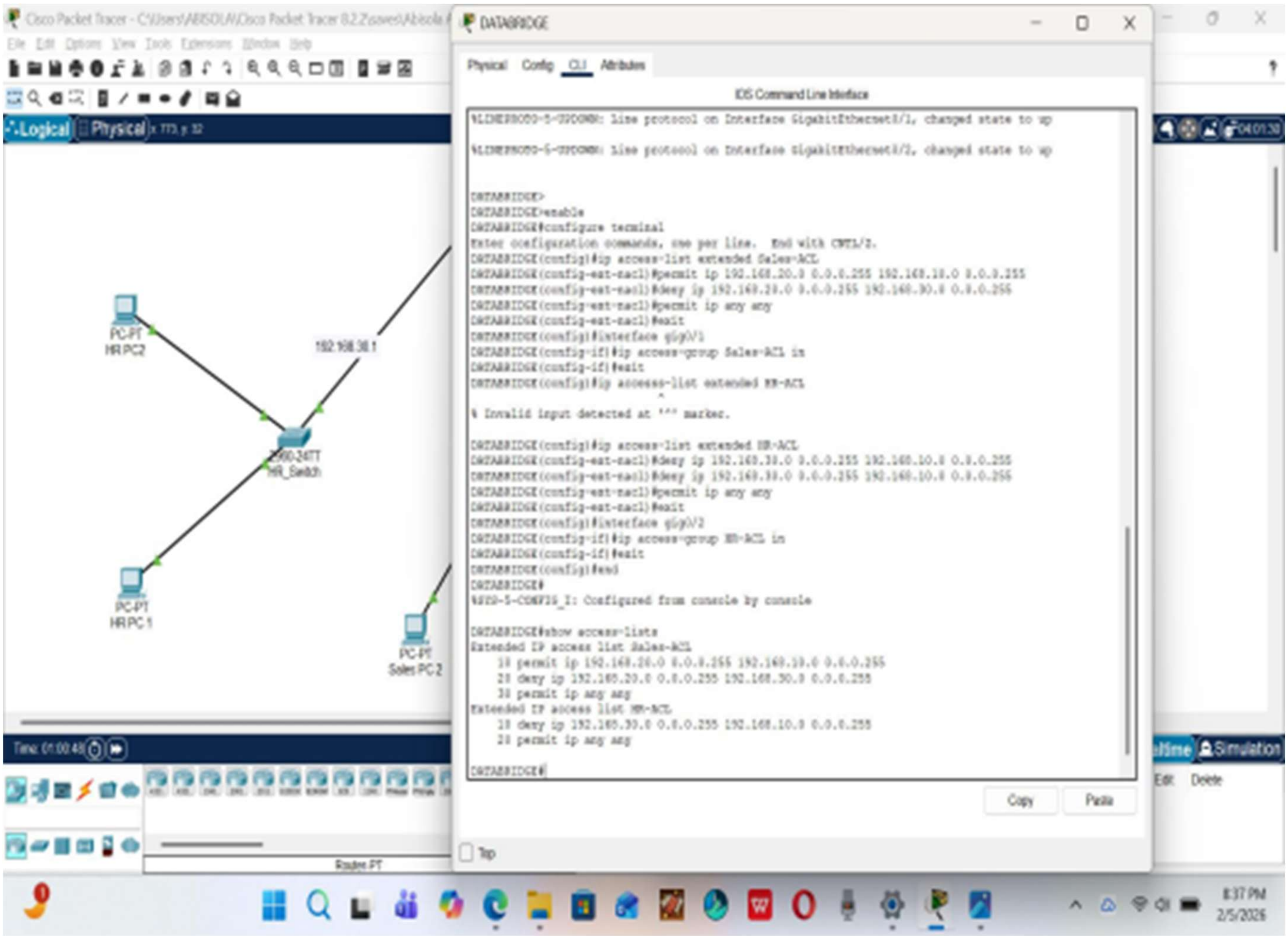



**Access-lists (ACLs) configuration** : This command was used to secure internal resourcesbyfiltering network traffic based on IP and protocol.The Sales subnet was permittedaccesstothe Admin server while the HR subnet was completely denied access. This ensuresthatunauthorized users cannot communicate/reach internal systems,which demonstrateshowACLs enhance network security and enforce departmental boundaries.

ACLs played a crucial role in securing network by controlling and filtering trafficflowbetween departments . ACLs effectively regulated communication paths, preventinglateralmovement and limiting exposure of sensitives areas.



Figure 4: This shows the access-lists (ACLs) command input and verificationof theoutput.




Verification of the ACLs rules applied : The command prompt mode was usedtoverifyifthe ACLs rules has been applied to show:

Failed ping/communication from HR-Dept to Admin- Dept.

Successful ping/communication from Sales-Dept to Admin -Dept.



Figure 5: This shows successful ping from Sales -Dept to Admin-Dept.

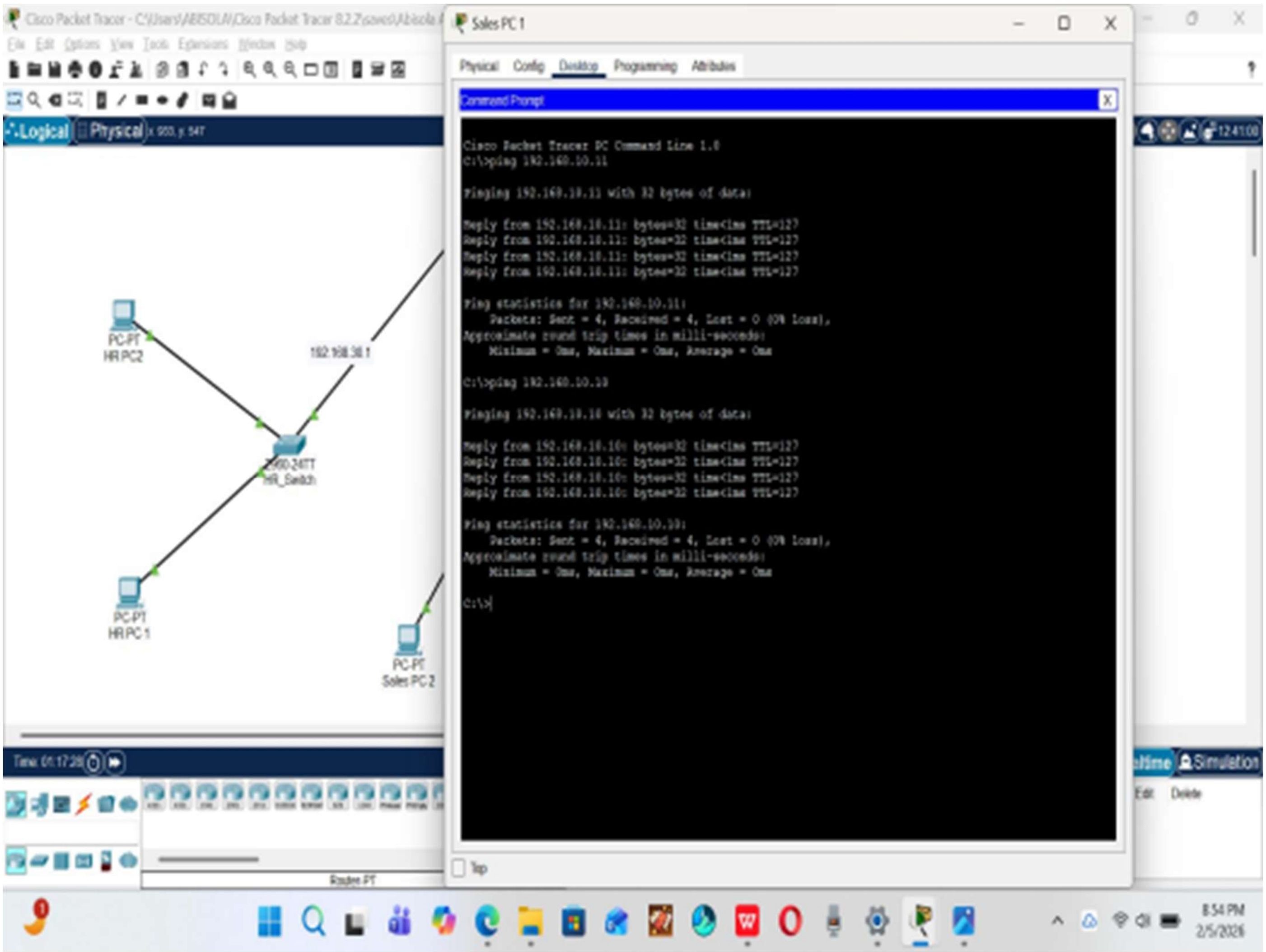
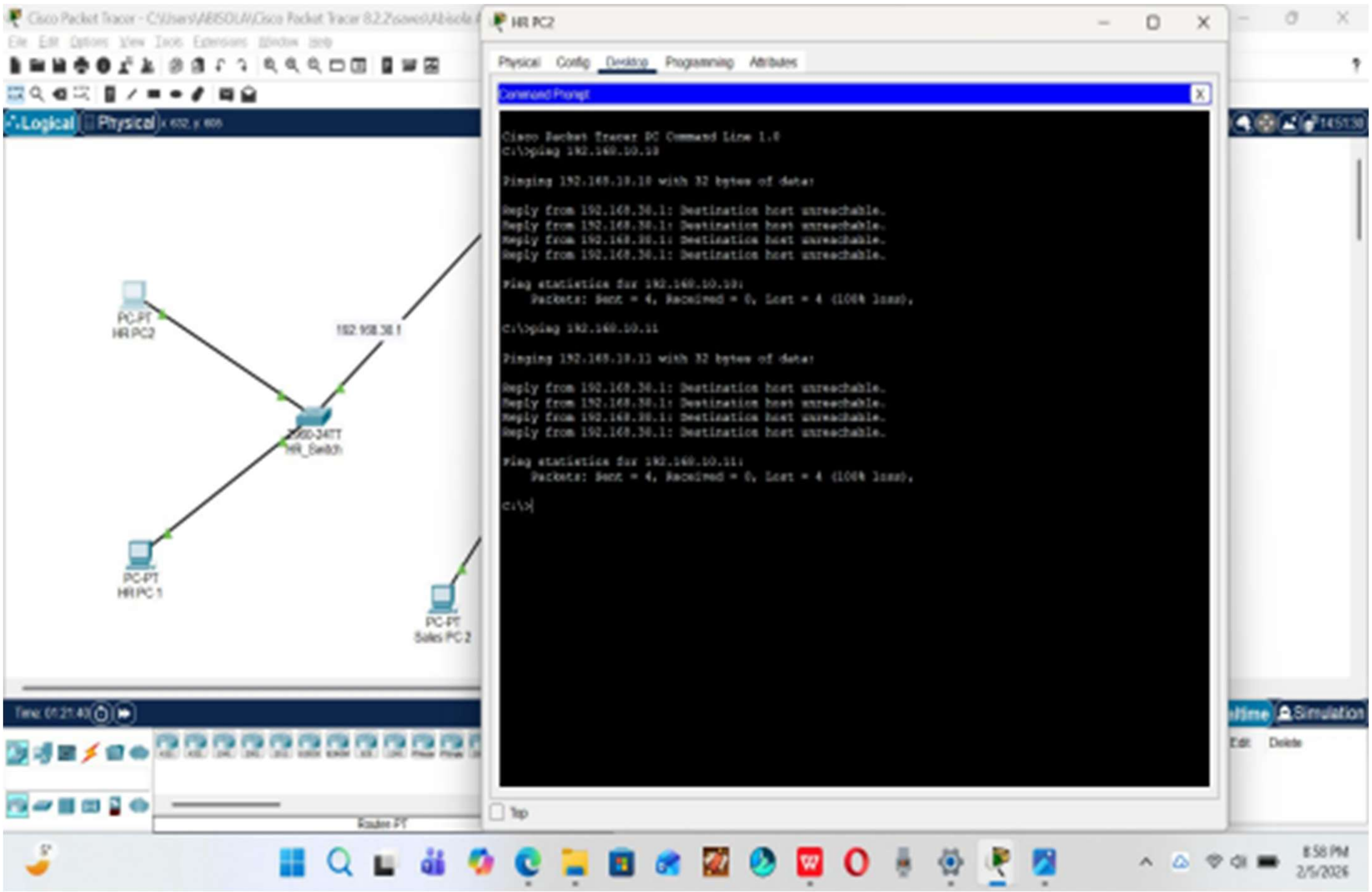





Figure 6: This shows the failed ping from HR to Admin




## Conclusion

ACLs is a fundamental tool in network management, enabling secure ,controlled, andefficient traffic flow. They protect resources, enforce policies and segment networks, balancing security and functionality.They enable network administrators to specifywhichusers or departments are permitted to access a certain network resources ,therebyensuringthat only legitimate connections are allowed. Moreover, ACLs help block authorizedaccess,minimize security threats, and protect stability of network operations.ACLs contributetoimprove network efficiency by managing traffic effectively and promote compliancebyaligning user access rights with established security standards and organizational policies.

In this project , Access control lists(ACLs) was configured on a router(Data-bridge) tomanage and secure traffic flow between departments ( Admin,Sales,HR) in a small networkthrough the CLI mode. Verification was done through the command prompt modetoverifythe successful and failed pings.